Numericals

i) For, $\beta = 90^\circ$:- (Vertical surface)

$$\boxed{\cos\theta = \sin\phi \cos\delta \cos\gamma \cos\omega - \cos\phi \sin\delta \cos\gamma + \cos\delta \sin\gamma \sin\omega}$$

ii) For, $\beta = 0^\circ$:- (Horizontal surface) $\Rightarrow$ sun at zenith.

$$\boxed{\cos\theta = \sin\phi \sin\delta + \cos\phi \cos\delta \cos\omega}$$

iii) For inclined surface facing due south :- $(\gamma = 0^\circ)$

$$\cos\theta = \sin\phi\,(\sin\delta \cos\beta + \cos\delta \cos\omega \sin\beta) + \cos\phi\,(\cos\delta \cos\omega \cos\beta - \sin\delta \sin\beta)$$

$$\Rightarrow \boxed{\cos\theta = \sin\delta \cdot \sin(\phi - \beta) + \cos\delta \cos\omega \cos(\phi - \beta)}$$

iv) Vertical surface facing due south :- $(\gamma = 0^\circ, \beta = 90^\circ)$

$$\cos\theta = \sin\delta \sin(\phi - 90^\circ) + \cos\delta \cos\omega \cos(\phi - 90^\circ)$$

$$\Rightarrow \boxed{\cos\theta = \sin\phi \cos\delta \cos\omega - \sin\delta \cos\phi}$$

v) For inclined surface facing due North :- $(\gamma = 180^\circ)$

$$\boxed{\begin{aligned}\cos\theta = {} & \sin\phi\,(\sin\delta \cos\beta - \cos\delta \cos\omega \sin\beta) \\ & + \cos\phi\,(\cos\delta \cos\omega \cos\beta + \sin\delta \sin\beta)\end{aligned}}$$

vi) Vertical surface facing due North :- $(\gamma = 180^\circ, \beta = 90^\circ)$

$$\boxed{\cos\theta = -\sin\phi \cos\delta \cos\omega + \cos\phi \sin\delta}$$

Problem 1

- Calculate the angle of incidence of beam radiation on a flat plate collector for
- Nagpur (21 ° 6 ‘ N, 79 °03 ‘ E)
- Slope of the collector = 10 °
- Surface azimuthal angle = 15 °
- Date = 1st Dec 1979
- Time 9 hr LAT

- Latitude angle = φ =
- Slope of collector = β
- Surface azimuthal angle = γ
- Hour angle = w
- n

- Latitude angle = φ =21° 06'=21.(06/60)=21.01 °
- Slope of collector = β=10 °
- Surface azimuthal angle = $\gamma = 15°$
- Hour angle = w= (180-(9x15))=45°
- n=31+28+31+30+31+30+31+31+30+31+30+01=335

$$\delta = 23.45 \text{x} \sin\left[\frac{360}{365}(284+n)\right] \text{degree}$$

$$= -22.10^{\circ}$$

$$\cos\vartheta_i = \begin{bmatrix} \cos\delta\cos\omega(\cos\phi\cos\beta + \sin\phi\sin\beta\cos\gamma) \\ +\cos\delta\sin\omega\sin\beta\sin\gamma \\ +\sin\delta(\sin\phi\cos\beta - \cos\phi\sin\beta\cos\gamma) \end{bmatrix}$$

$$\vartheta_i = 53.34^{\circ}$$

Problem 2

- Calculate the angle made by beam radiation with the normal to a flat plate collector on May 1st at 9 hr (LAT). The collector is located in New Delhi (28°35'N,77°12'E).It is tilted at an angle of 36° with the horizontal and is pointed due south. Repeat the calculation for 12 hr.

- Latitude angle = φ =
- Slope of collector = β
- Surface azimuthal angle = γ
- Hour angle = w
- n

- Latitude angle = φ =28° 35'=28.(35/60)=28.58 °
- Slope of collector = β=36 °
- Surface azimuthal angle = $\gamma = 0°$
- Hour angle = w= (180-(9x15))=45°
  - w= (180-(12x15))=0°

- n=31+28+31+30+1=121

$$\delta = 23.45 \text{x} \sin\left[\frac{360}{365}(284+n)\right] \text{degree}$$

$$= 14.9^{\circ}$$

$$\cos \vartheta_i = \begin{bmatrix} \cos\delta\cos\omega(\cos\phi\cos\beta + \sin\phi\sin\beta\cos\gamma) \\ +\cos\delta\sin\omega\sin\beta\sin\gamma \\ +\sin\delta(\sin\phi\cos\beta - \cos\phi\sin\beta\cos\gamma) \end{bmatrix}$$

$$\vartheta = 49.87^{\circ}, 22.32^{\circ}$$

Problem 3

- Calculate the declination angle of beam radiation on a plane surface, tilted by 45° from horizontal plane and pointing 30° west of south located at Mumbai at 1:30 PM (LAT) on 18° 54' N on 15th November. The longitude and latitude of Mumbai are 72° 49' E and 18° 54’ N respectively.

$$\delta = 23.45x \sin\left[\frac{360}{365}(284+n)\right] \text{degree}$$

$$n = 319 (\text{date} = 15^{th} \text{ november})$$

$$\delta = -19.148^{\circ}$$

Problem 4

Calculate the declination angle of beam radiation on a plane surface, tilted by 45° from horizontal plane and pointing 30° west of south located at Mumbai at 1:30 PM (LAT) on 18° 54' N on 15th November. The longitude and latitude of Mumbai are 72° 49' E and 18° 54′ N respectively.

$\gamma=30^\circ, \beta=45^\circ, \phi=18.9^\circ$

$$\cos\vartheta_i=\left[\begin{array}{l}\cos\delta\cos\omega(\cos\phi\cos\beta+\sin\phi\sin\beta\cos\gamma)+\cos\delta\sin\omega\sin\beta\sin\gamma\\+\sin\delta(\sin\phi\cos\beta-\cos\phi\sin\beta\cos\gamma)\end{array}\right]$$

$\vartheta=37.23^\circ$

Life cycle costing

- Life cycle costing (LCC) is a way to calculate the total cost of a solar photovoltaic (PV) system over its entire lifespan.
- It includes all costs, from the initial investment to the cost of maintenance and disposal.

- **Capital costs**: The cost of the initial investment, including the cost of the panels, electrical equipment, and mounting structure

- **Operational costs**: The cost of running the system, including maintenance and repairs

- **Energy costs**: The cost of the energy used by the system

- **Disposal costs**: The cost of safely disposing of the panels at the end of their life

LCC:-

i) It stands for Life Cycle Costing.

ii) It is defined as an important factor for comparing the alternatives and deciding on a particular process for completing a project.

iii) The different components taken into account for calculati LCC are:-

$$\boxed{LCC = \text{Capital Cost} + \text{Replacement cost} + \text{Maintanance cost} + \text{Energy cost} - \text{Salvage}.}$$

∴ Calculate the maintanance cost.

$$\boxed{M_{\text{present worth}} = M_{\text{per yr}} \times \frac{1}{i}\left[1 - \left(\frac{1}{1+i}\right)^n\right]}$$

A PV-array of 500 watts has been installed to pump water from a borewell of 2m deep using a submergable motor & pump system to an overhead tank. The length of pipe requir to pump the water is 30 m. Following are the cost involved f the subsystem & their life spans :-

- Pv array=8/watt=life span 15yrs
- Motor and pump cost=2/watt=life span=7.5rs
- Pipe cost=8/m=lifespan=5yrs
- Cost of digging=20/m
- Maintainace cost=80/yr
- Miscellaneous cost=3.5/watt
- If the rate of interest is 10 % calculate the LCC of 15yrs.

Ans:- Given,

$i = 10\%$, $n = 15$ yr

↳ i) Capital cost:-

Capital cost = cost of PV array + cost of Motor & Pump + Pipe cost + cost for digging the borewell

$\Rightarrow C_C = C_{PV} + C_M + C_P + C_B$

Here,

$C_{PV} = 500 \times 8\$ = (4000\$) = \$4000$

$C_M = 500 \times 2\$ = \$1000$

$C_P = 30 \times 8\$ = \$240$

$C_B = 2 \times 20\$ = \$40$

So,

$C_C = 4000 + 1000 + 240 + 40 \Rightarrow \boxed{C_C = \$5280}$

↳ <u>Replacement Cost:-</u>

Here,

$$R_C = R_M + R_{pipe}$$

Calculate the maintanance cost.

$$\boxed{M_{present\ worth} = M_{per\ yr} \times \frac{1}{i}\left[1-\left(\frac{1}{1+i}\right)^n\right]}$$

Now,

$$R_M = \frac{1000}{(1+0.1)^{7.5}} = \$\,489.27$$

$$R_{pipe} = \frac{240}{(1+0.1)^5} = \$\,149.02 \longrightarrow \text{for 1st replacement}$$

$$= \frac{240}{(1+0.1)^{10}} = \$\,92.53 \longrightarrow \text{for 2nd replacement}$$

So,

$$R_C = R_M + R_{pipe} = (489.27 + 149.02 + 92.53)$$

$$\Rightarrow \boxed{R_C = \$\,730.82}$$

↳ iii) <u>Maintanance cost :-</u>

Given, Maintanance cost/yr = M_c/yr = \$ 80/yr

We know,

$$M_{c\,present} = M_c/yr \times \frac{1}{i}\left[1-\left(\frac{1}{1+i}\right)^n\right]$$

$$\Rightarrow M_{c\,present} = 80 \times \frac{1}{0.1}\left[1-\left(\frac{1}{1.1}\right)^{15}\right]$$

$$\Rightarrow \boxed{M_c = \$608.48}$$

Now, we know that,

$$LCC = C_c + R_c + M_c + (\text{Miscellaneous cost})$$

$$\Rightarrow LCC = \$(5280 + 730.82 + 608.48) + [\$3.5/\text{watt}]$$

$$\Rightarrow LCC = \$(5280 + 730.82 + 608.48) + [\$3.5 \times 500]$$

$$\Rightarrow \boxed{LCC = \$8368.486}$$